KENNESAW STATE
UNIVERSITY

CS 4277
Introduction to Deep Learning

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Computer Science

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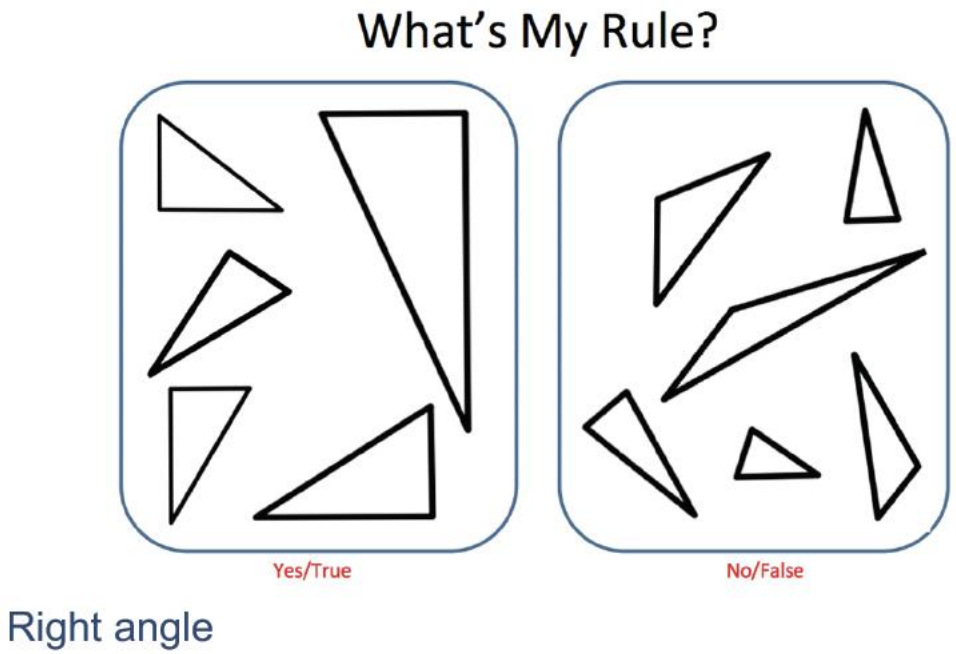
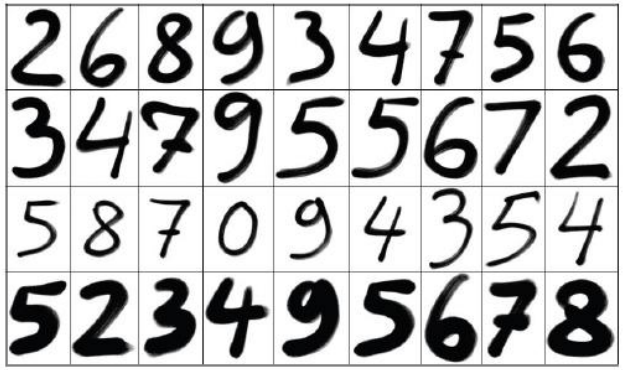
How to develop an artificial intelligence system?

- Expert knowledge (manual rules)



What is the Rule?



Machine learning →
find the rules

Machine learning ≈ build a mapping function

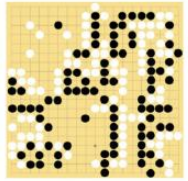
Speech Recognition

$f($ $)=$ “Hello”

Image Identification

$f($ $)=$ “9”

Go

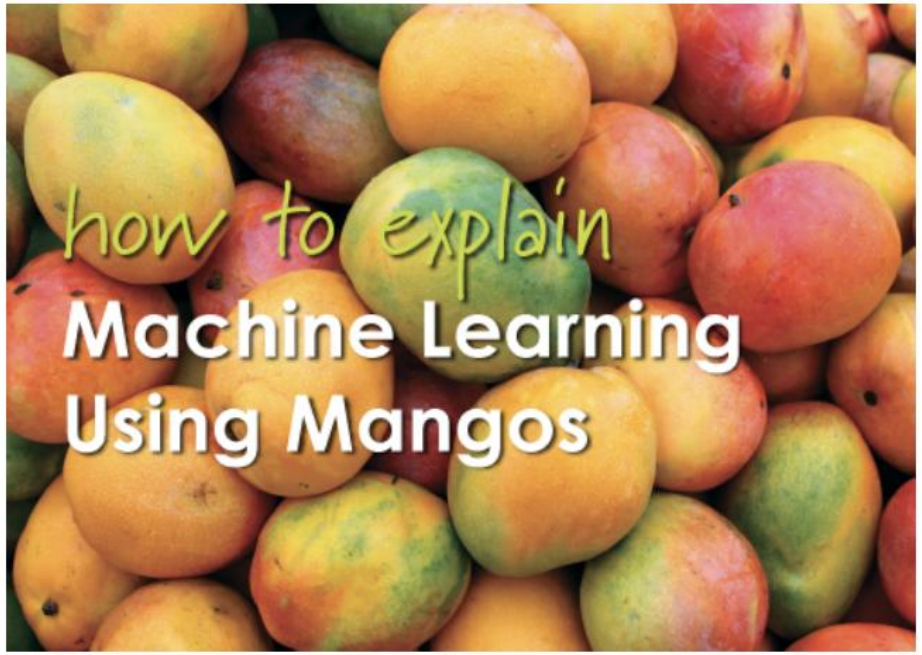
$f($  $)=$ “6-5” (position)

Machine translation

$f($ “你好！” $)=$ “Hello!”

Example: Mango machine learning

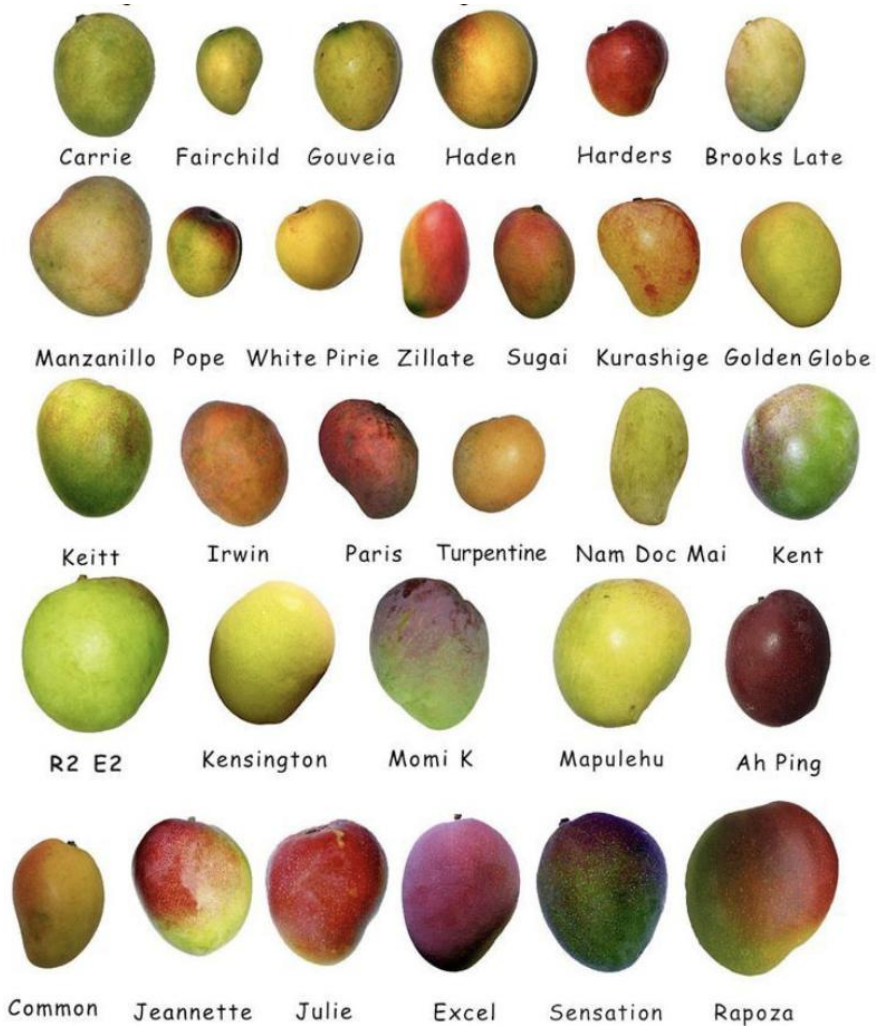
Q: How to determine whether mangoes are sweet?



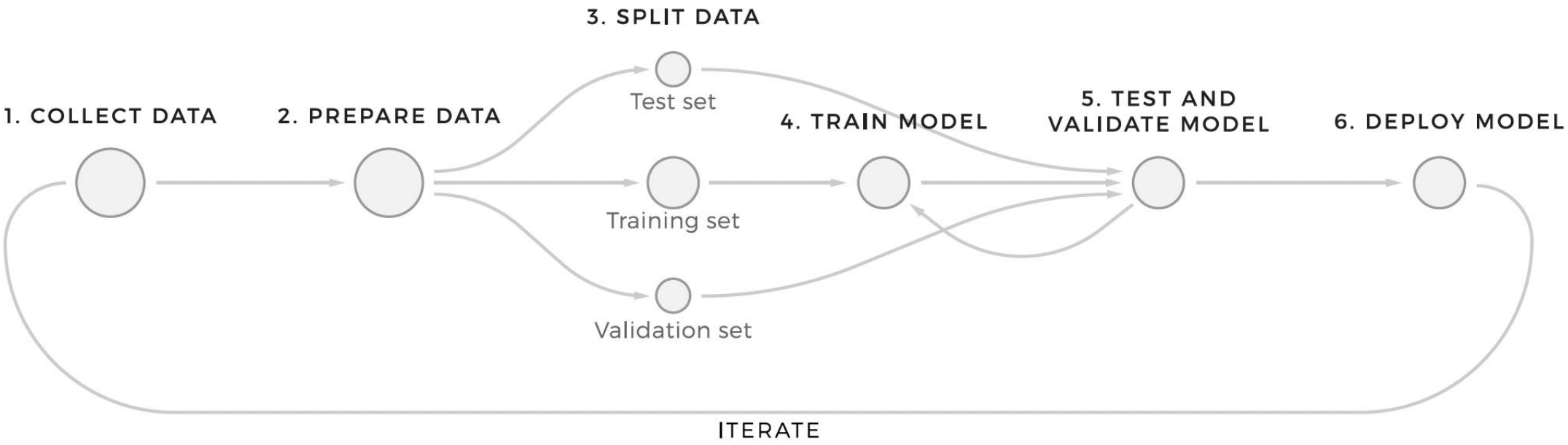
https://www.quora.com/How-do-you-explain-Machine-Learning-and-Data-Mining-to-non-Computer-Science-people

Mango machine learning

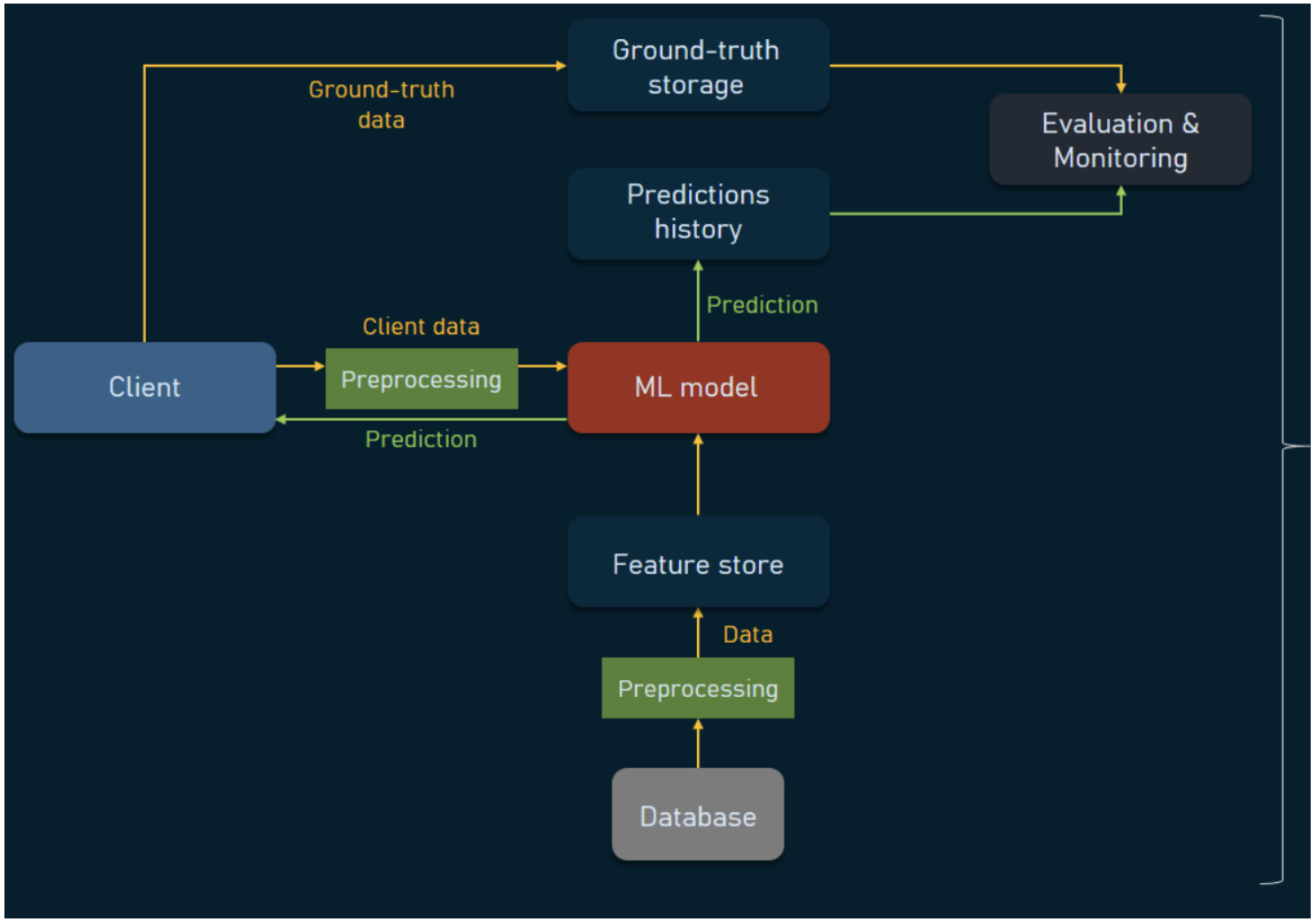
- From a randomly selected mango sample (training data) from the market, list all the characteristics of each mango:
  - Such as color, size, shape, origin, brand
- And mango quality (output variable):
  - Sweet, juicy, ripeness.
- Design a learning algorithm to learn the correlation model between mango characteristics and output variables.
  - f(input) = output
- Next time you buy mangoes from the market, you can use the previously calculated model to predict the quality of mangoes based on the characteristics of the mangoes (test data).



Machine learning workflows



Machine Learning Pipeline in production



Tasks Machine Learning Solves

- Five groups of task
  - Classification
  - Cluster analysis
  - Regression
  - Ranking
  - Generation

Classification

- Classification algorithms define which category the objects from the dataset belong to.
  - Binary class problem
  - Multiclass problem

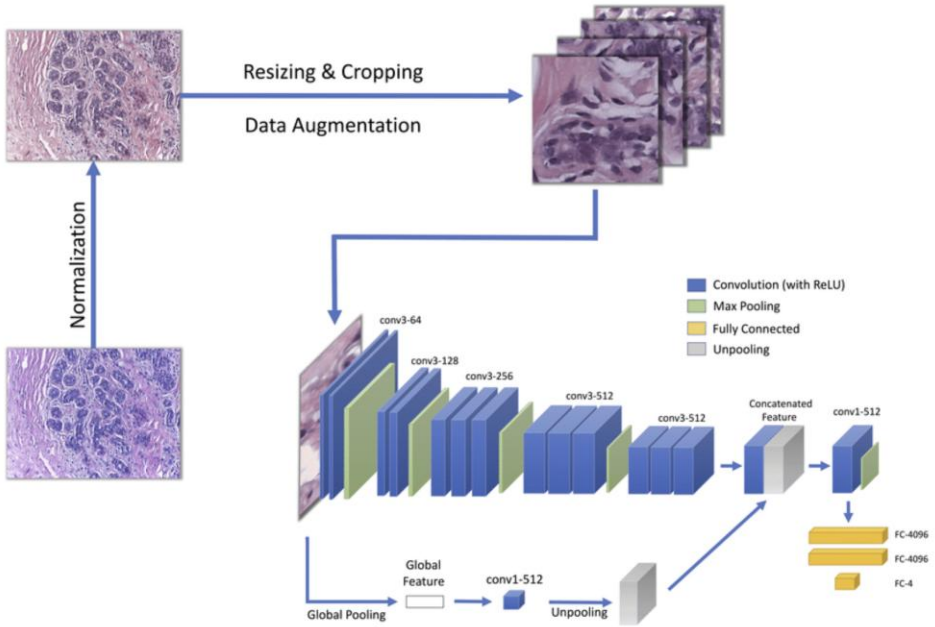
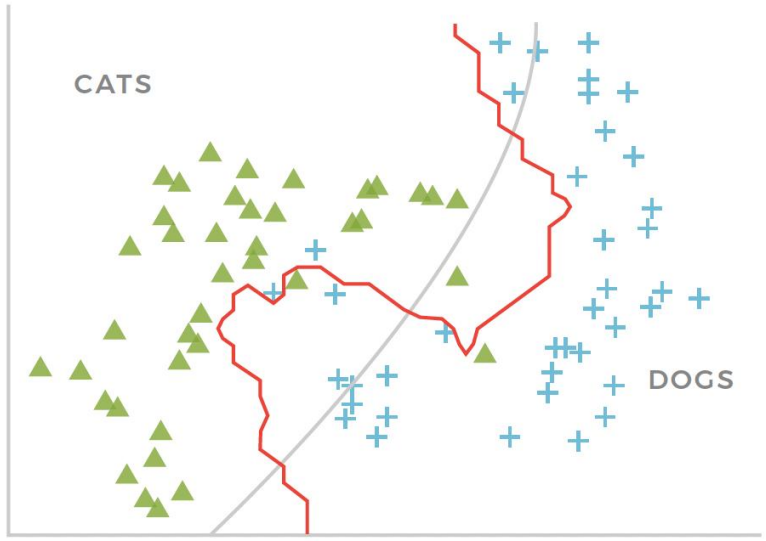
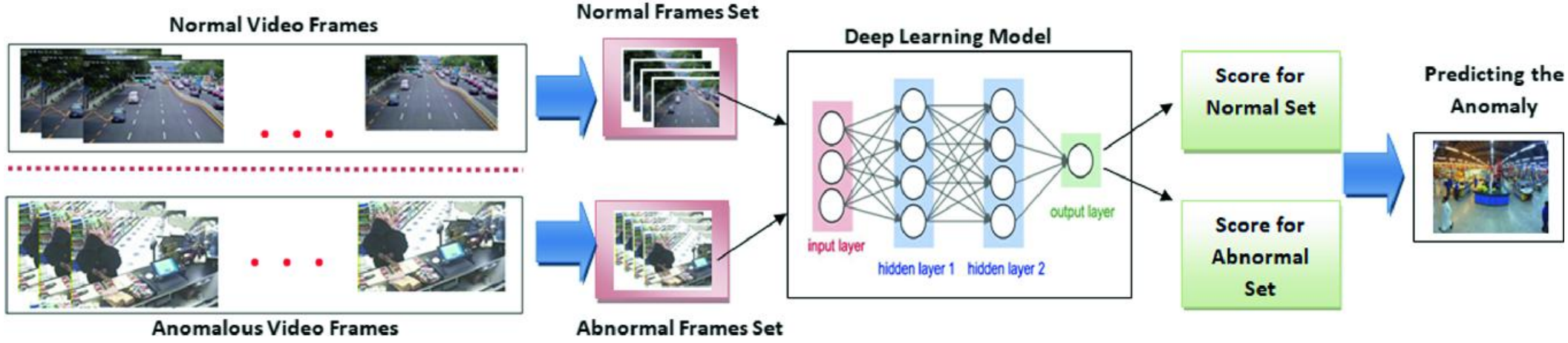


Illustration of Classification Pipeline



Binary classification

Classification – anomaly detection



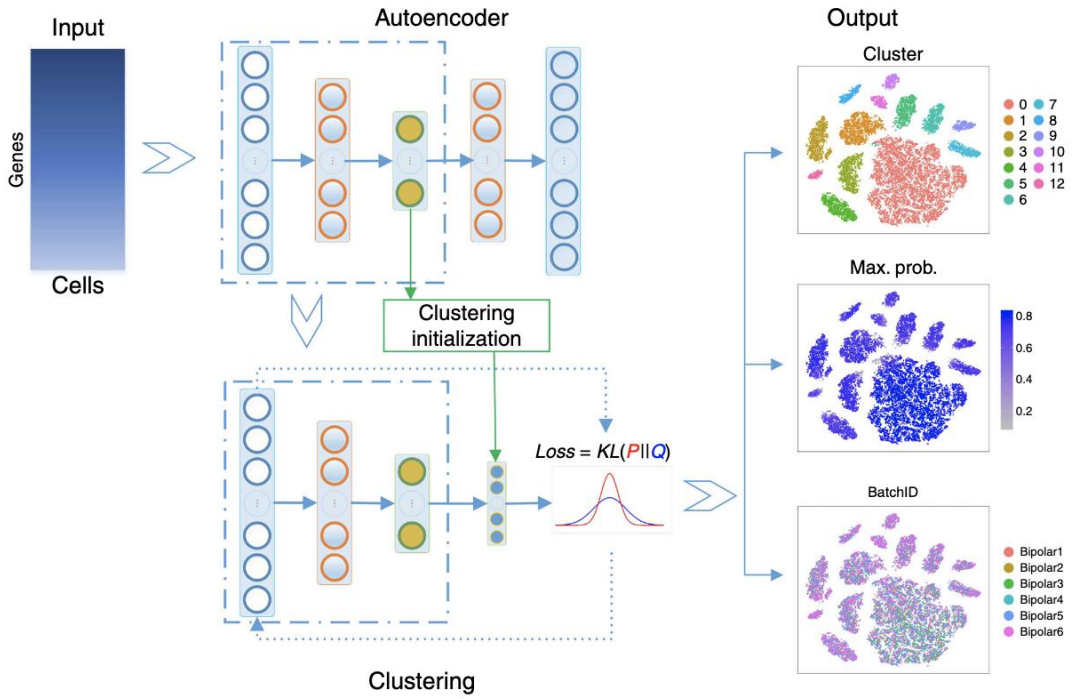
- The goal of anomaly detection is to find outliers, unusual objects in data that don't appear in its normal distribution.

Anomaly detection

Cluster analysis

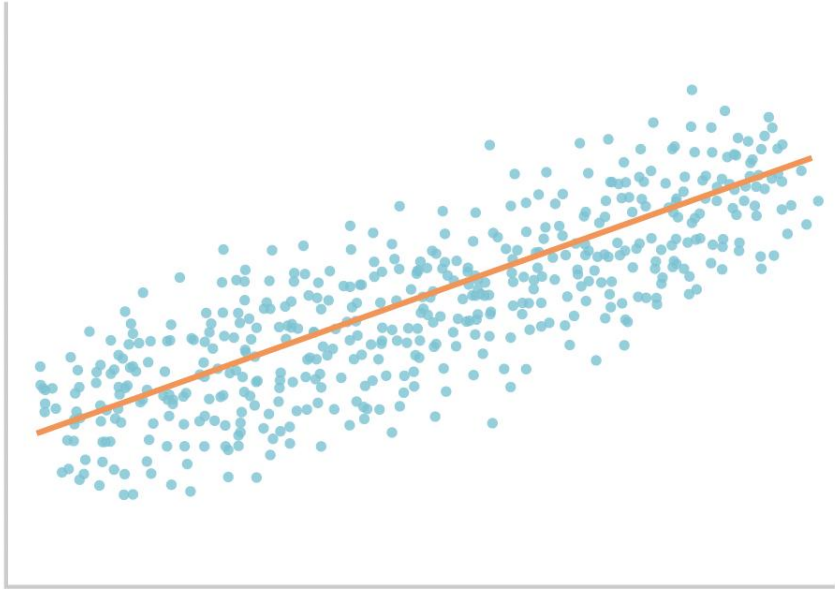
- The main difference between regular classification and clustering is that the algorithm is challenged to group items in clusters without predefined classes.

Example: Single-cell RNA sequencing (scRNA-seq) can characterize cell types and states through unsupervised clustering



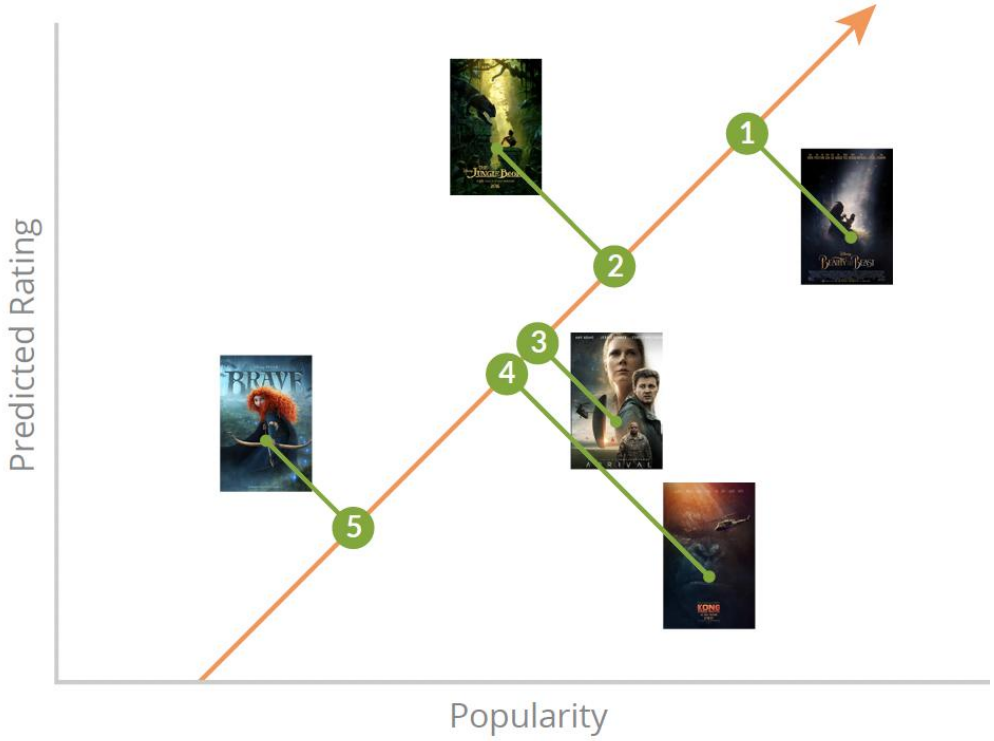
Regression Analysis

- Regression algorithms define numeric target values instead of classes. By estimating numeric variables, these algorithms are used in predicting product demand, sales figures, marketing returns, etc.
  - How many hours will this surgery take?
  - How much rainfall will this town have in the next six hours?
  - What's will the airfare be for this destination?



Ranking

- Ranking algorithms decide the relative importance of objects (or items) as related to other objects. The most well-known example is PageRank, which is heavily used by Google to rank pages on the search engine results page.
  - Which movies this user will enjoy the most?
  - What hotels will be on the most-recommended list for this customer?



Movie recommendation ranking

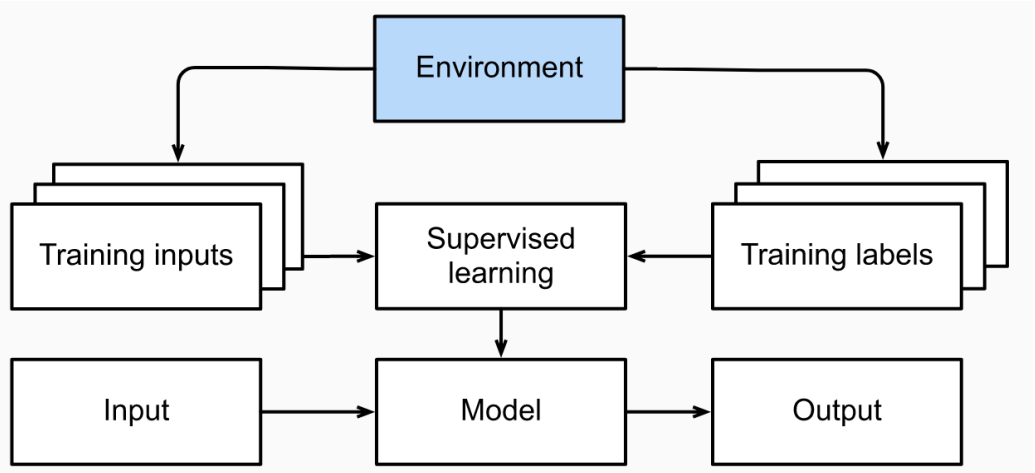
Generation

- Generation algorithms are applied to generate text, images, or music. Today they are used in such applications as Prisma that converts photos to artwork-style images, or **WaveNet by DeepMind** that can mimic human speech or create musical compositions.
- Turn photos into specific style of painting.
- Create text-to-speech applications for mobile
- Voice assistants (e.g. the Google assistant).
- Create music samples of one style or that are reminiscent of a particular musician.

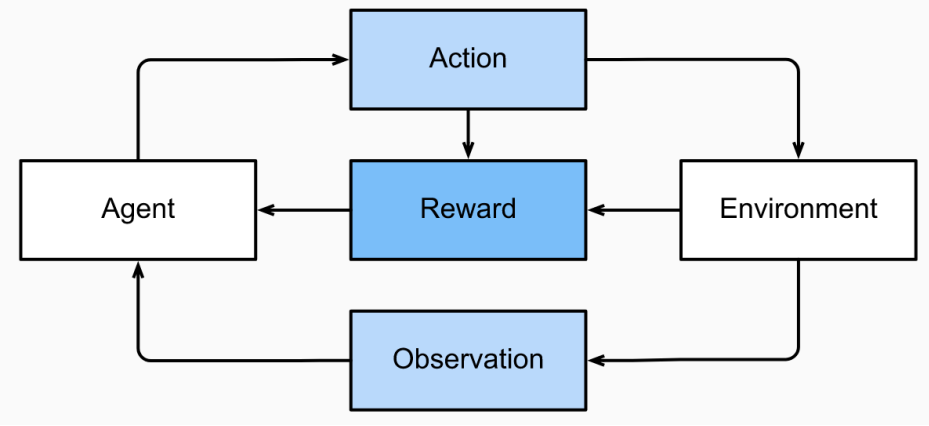
Image converted to artwork using "The Great Wave off Kanagawa" piece of art

Three Model Training Styles

- **Supervised learning**
  - Supervised learning algorithms operate with historic data that already has target values. Mapping these target values in training datasets is called labeling.
- **Unsupervised learning**
  - The goal of machine learning, in this case, is to define patterns in values and structure the objects by similarities or differences.
- **Reinforcement learning**
  - An agent (an algorithm) must make decisions based on input data and then be "awarded" or "punished," depending on how successful these decisions were.

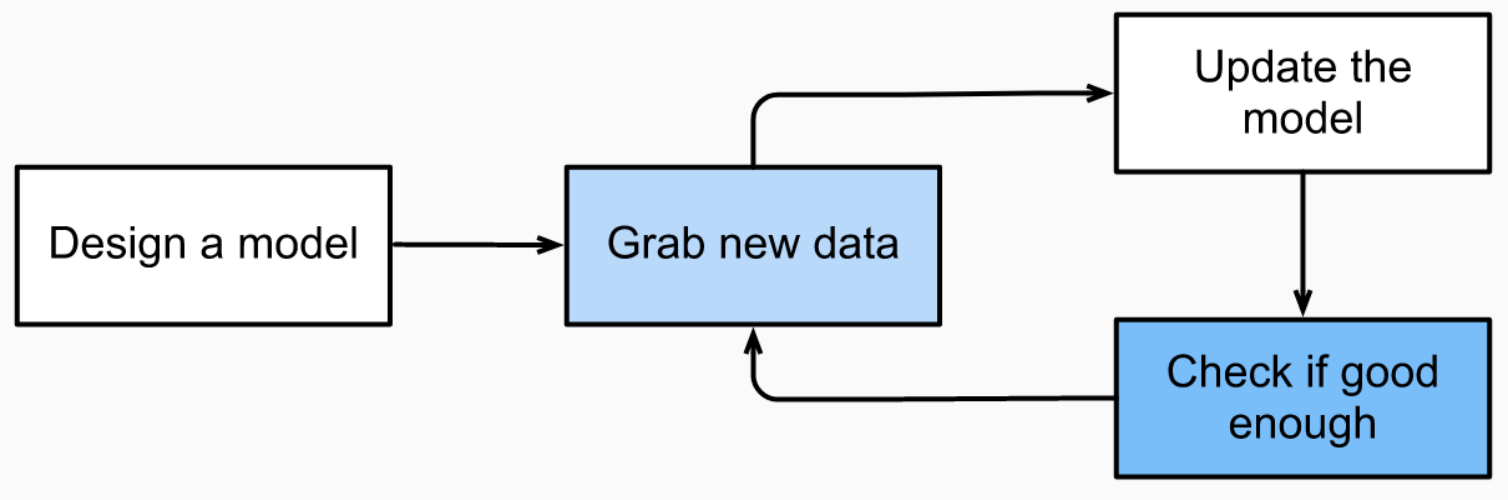


Supervised Learning



Reinforcement Learning

Components in Supervised Training



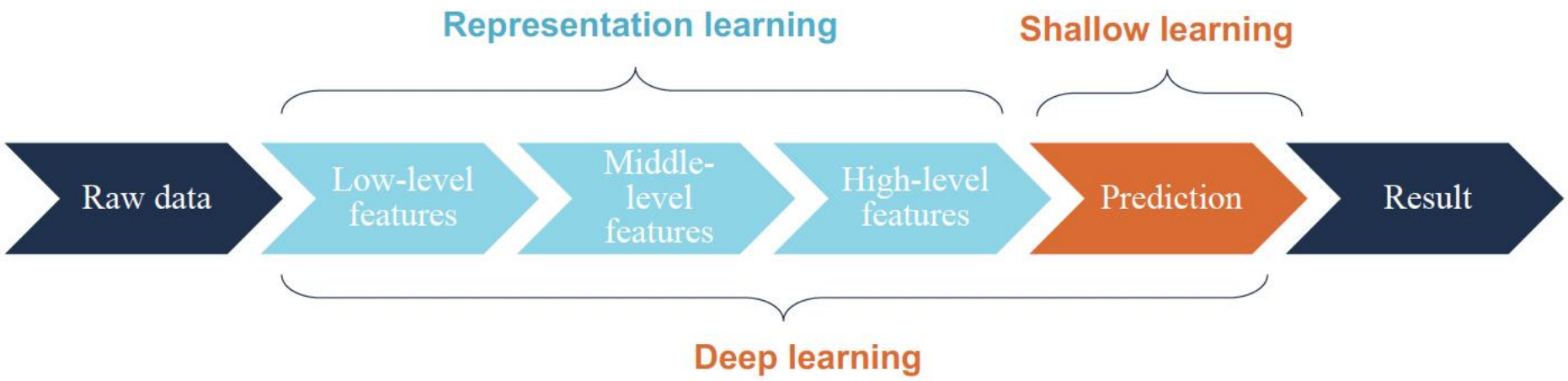
- **Key Components:**
- The ***data*** that we can learn from.
- A ***model*** of how to transform the data.
- An ***objective function*** that quantifies how well (or badly) the model is doing.
- An ***algorithm*** to adjust the model's parameters to optimize the objective function.

Types of Supervised Models

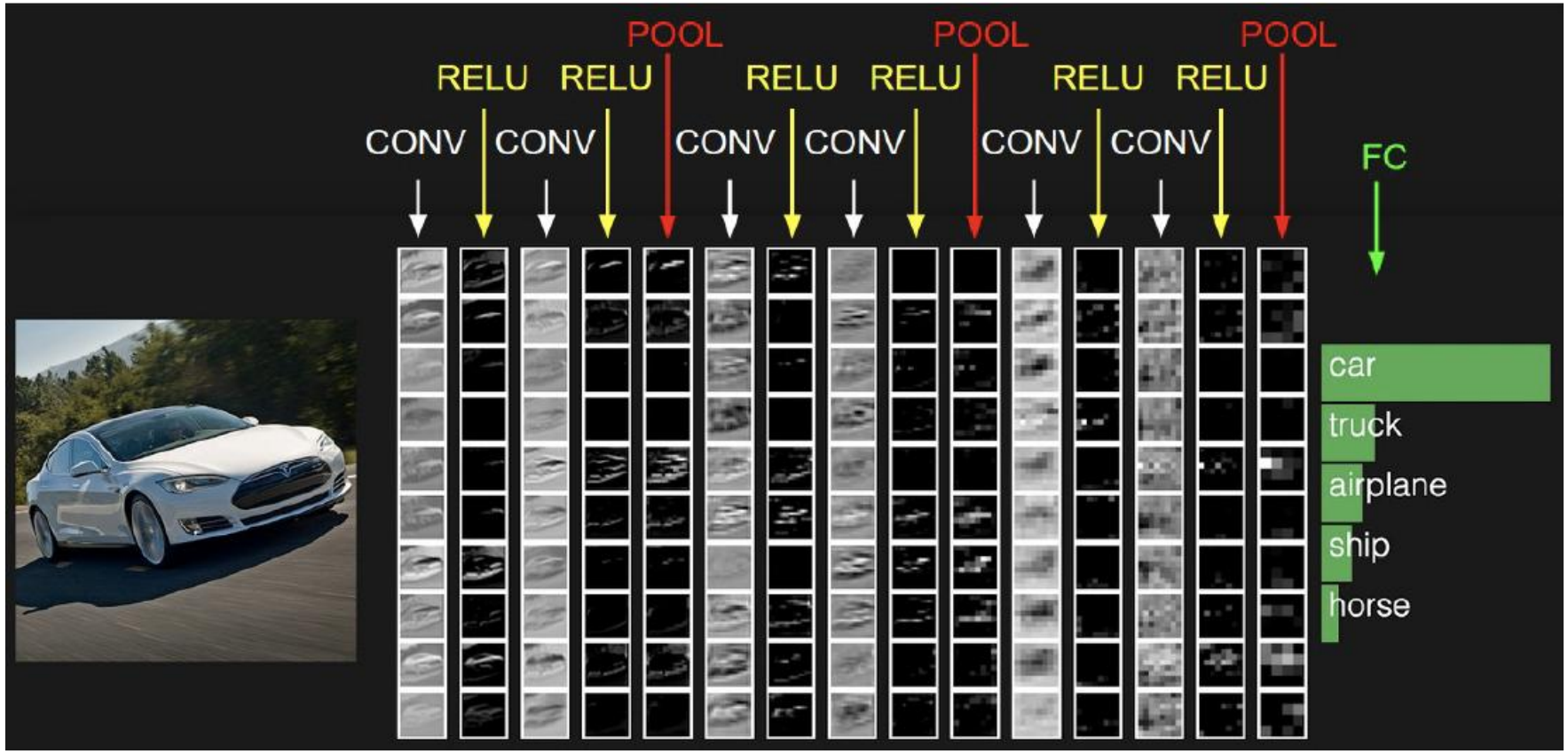
Decision trees	• Use trees to make decisions
Linear methods	• Decision is made from a linear combination of input features
Kernel machines	• Use kernel functions to compute feature similarities
Neural Networks	• Use neural networks to learn feature representations

Deep learning

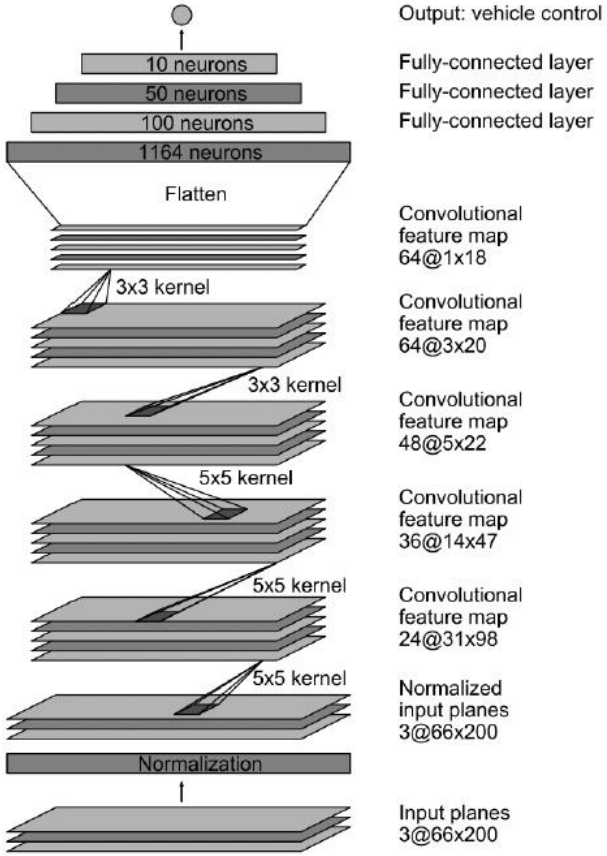
- By building a model with a certain "depth", the model can automatically learn a good feature representation (from low-level features, to middle level features, and then to high-level features), thereby ultimately improving the accuracy of prediction or recognition



Deep learning: image recognition



Deep learning: self driving car



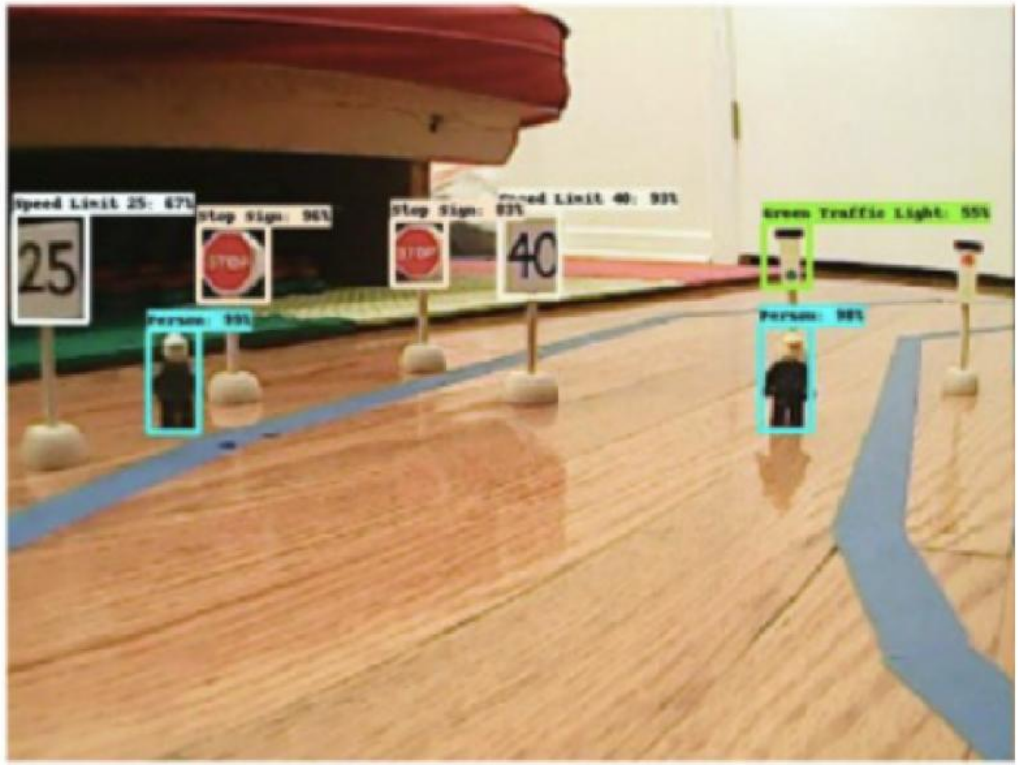
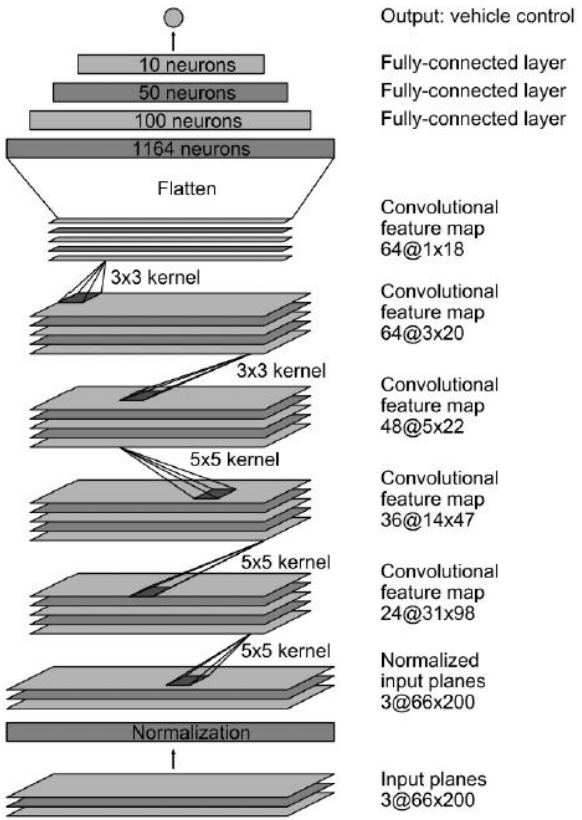
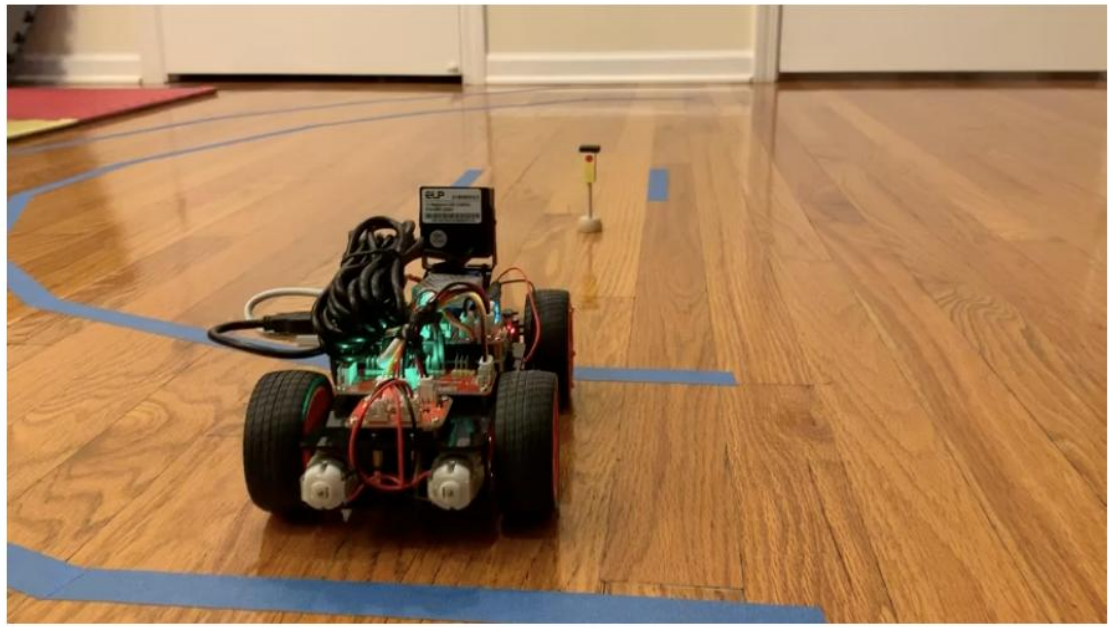


Figure 4: CNN architecture. The network has about 27 million connections and 250 thousand parameters.

Deep learning: self driving car





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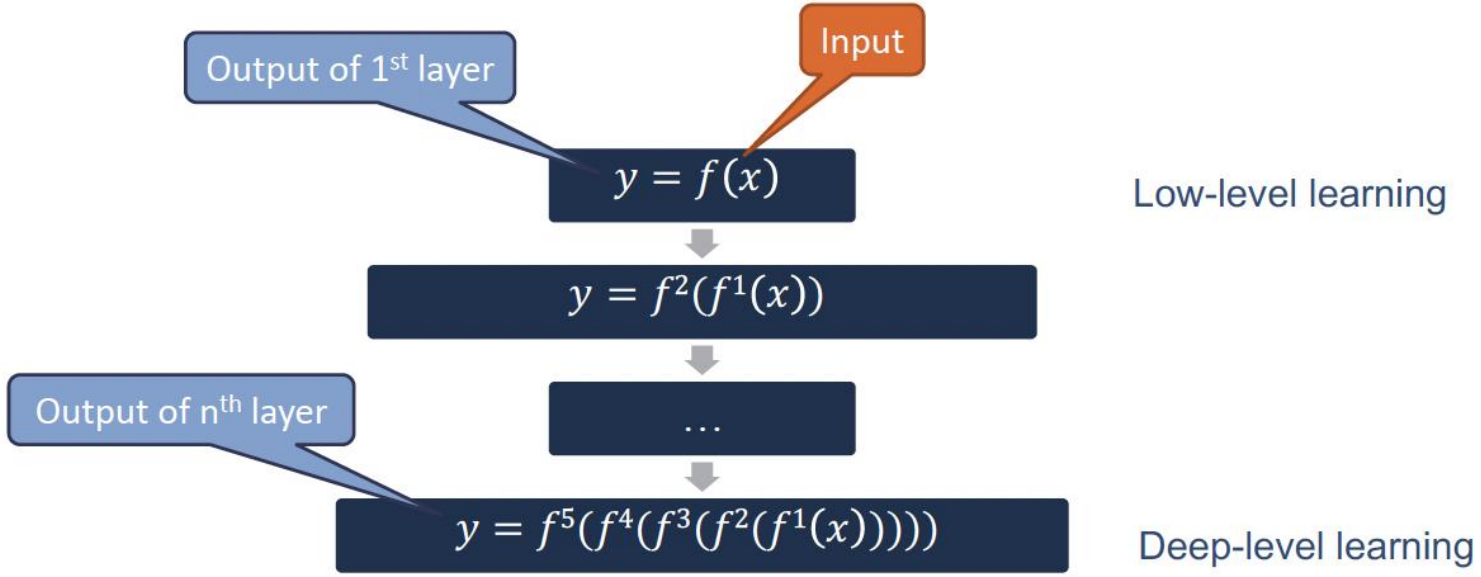
Mathematical description of deep learning



$f^l(x)$ *is a nonlinear function, not necessarily continuous.*